

MCA (Revised) / BCA (Revised)

Term-End Examination

June, 2021

**MCS-011 : PROBLEM SOLVING AND
PROGRAMMING**

Time : 3 hours

Maximum Marks : 100

(Weightage : 75%)

Note : *Question no. 1 is **compulsory**. Answer any **three** questions from the rest.*

1. (a) Write an algorithm and draw a corresponding flow chart to check whether the given year is a leap year or not. 10
- (b) Write a C function **isodd(num)** that accepts an integer argument and returns 1 if the argument is odd and 0 otherwise. 4
- (c) Write a C program that invokes this function to generate numbers between the given range. 6

- (d) Define an Array. How do we declare and initialize a single-dimensional array and a 2-dimensional array ? Explain with an example for each. 4
- (e) Write a C program to implement STRING COPY operation that copies a string “**str1**” to another string “**str2**” without using library function. 8
- (f) Write a C program to maintain a record of “n” student details using an array of structures with structure-variables (Rollnumber, Name, Marks1, Marks2, Marks3, Marks4 and Grade). Define each field of the structure with appropriate datatype. Print the marks of the student of given Rollnumber as input. 8
- 2.** (a) Discuss dynamic memory allocation in C. Write and explain dynamic allocation functions in C. 6
- (b) Write and explain any two pre-processor directives in C. 4
- (c) Write a C program to find the sum of diagonal elements of a 3×3 matrix. 10
- 3.** (a) Write a C program to create a file of numbers given by the user and copy odd numbers to odd.dat file and even numbers to even.dat file. 10
- (b) *Using recursion*, write a C program to find the factorial of a given number. 10

4. (a) Write a function to print the sum of the following series :

$$1 + 2^2 + 3^3 + \dots + n^n$$

where “n” is passed as an argument to the function.

8

- (b) Write a program to calculate the number of vowels (a, e, i, o, u) separately in the entered string and display their individual count.

12

5. (a) Write two differences between a structure and a union with appropriate examples.

4

- (b) Write a C program to find the substring in a string without using a library function.

10

- (c) Write a program to delete an element from a given location of an array of integers.

6
